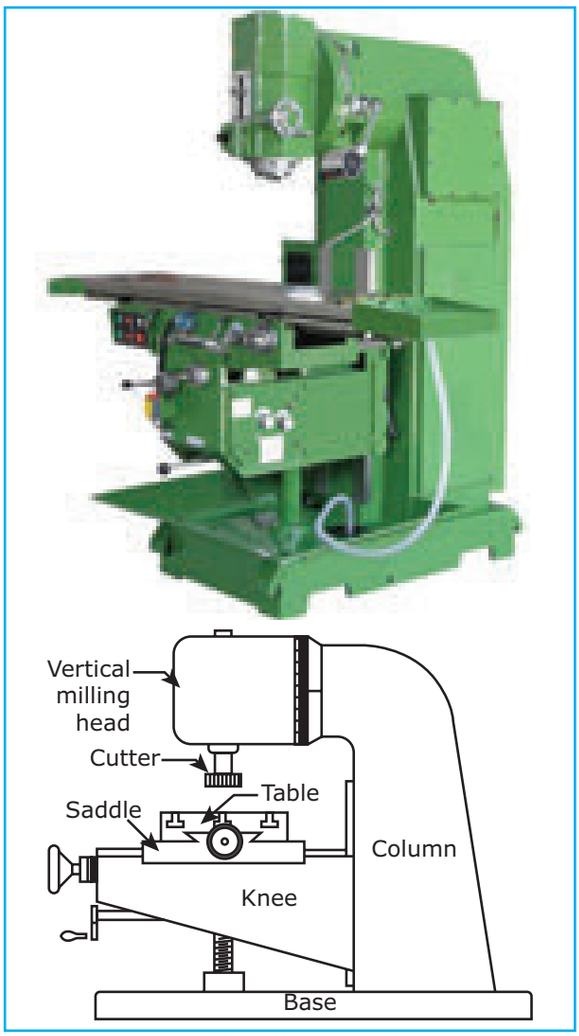
A column and knee type milling machine is illustrated in fig.

5.3 Vertical milling machine

It is very similar to a horizontal milling machine in construction as it has the same parts of base, column, knee, saddle and table. The spindle of the machine is positioned vertically. The cutters are mounted on the spindle is rotated by the power obtained from the mechanism placed inside the column. Angular surfaces are machined by swiveling the spindle head.

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Vertical milling machine

5.4 Types of milling machine

The milling machines are classified according to the general deign of the machine.

1. Column and knee type
   a. Plain milling machine
   b. Universal milling machine
   c. Omniversal milling Machine
   d. Vertical milling machine
2. Table types milling machine
3. Planer type milling machine
4. Special type milling machine

Column and knee type milling machine

The column of a column and knee type milling machine is mounted vertically upon the base. Knee is mounted on the accurately machined guideways of the column. It is designed to move up and down accurately. Saddle and table are mounted on the knee.

There are different types of column and knee type machines.

a. Plain milling machine

It is rigid and sturdy. Heavy workpieces are mounted and machined on the machine. The work mounted on the table is moved vertically, longitudinally and crosswise against the rotating cutter. The table cannot be rotated. It is also called as horizontal milling machine because the cutter rotates in horizontal plane.

b. Universal milling machine

The table of a universal milling machine can be swiveled by 45^0 on either side and so helical milling works can be performed.

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